Running Case (HybridQA): *What was the original name of the city with an AFL club formed in 1989?"* **Target Ground Truth:** `"York"` • **Modality Fusion:** Table (`AFL_Ontario_0`) + Wikipedia Bridge (`/wiki/Toronto`)

Stage 1 (Sec. 3.1): Multi-Source Ingestion

Table: `AFL_Ontario_0`

Club	Formed	City	height
1989	Toronto	Tor. Eagles	
Hamilton	Ham. Tigers	1991	
height			

→ Matches constraint: **1989**

Wikipedia Bridge Passage: `Toronto` → `Doc:Toronto`

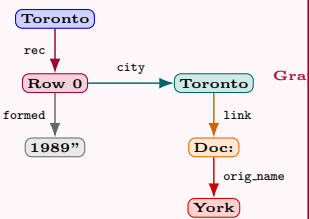
...capital of Ontario. Established in 1793 as to...

Retriever Output

• Candidate table rows • Linked Wikipedia passages • Complete context retained

Evidence

Stage 2 (Sec. 3.2): Multi-graph Construction



Graph + Path

Star Record Hubs

• Hub binds every cell to its row • Prevents cross-row conflation • Rule-based text links (no LLM)

Guided Reasoning Path

• Best path: Eagles → Row 0 → Toronto → York • Guides model attention (no filtering) • Compact serialized prompt

Stage 3 (Sec. 3.3): Two-Tier Controlled Engine

Tier 1: Primary Reader [LLM Call 1]

• **Prompt Packaging**: Compressed graph (deduplicated passages + row blocks) + guided reading path • **Controlled Decision**: Fast Exit Answer (1 call) if clear evidence; Diagnostic Handoff if ambiguous/missing

if collision

Diagnostic Handoff: `{"focus": ["Row 0", "Row 1"], "check": "original name"}`

Targeted Bridge Retrieval [0 LLM Calls]: Fetches missing passages for disputed rows; builds focused capsule

Focused Capsule

Tier 2: Targeted Reader [LLM Call 2]

• **Prompt Packaging**: Transmits *only missing & disputed facts* (discards all unrelated rows) • Resolves candidate ambiguity via verified facts ($\leq$ 2 calls total)

Fast Exit

Tier 2 Verified

Stage 4 (Sec. 3.4): Grounded Provenance

Verified Answer Span

`"York"`

• Exact context-grounded extraction • Zero hallucination

Auditable Provenance

• Table: `AFL_Ontario_0` • Row: `Row 0` (`Tor. Eagles`) • Cell: `City: Toronto` • Link: `/wiki/Toronto` • Text: *...town of York..."*

Reverse Grounding Fallback

• Maps answer back to graph coordinates • Direct synthesis if disconnected • Auditable evidence trail